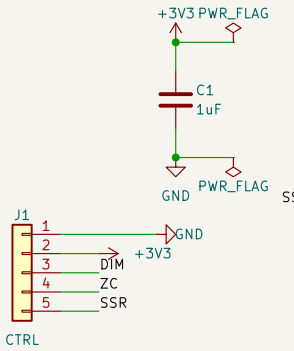
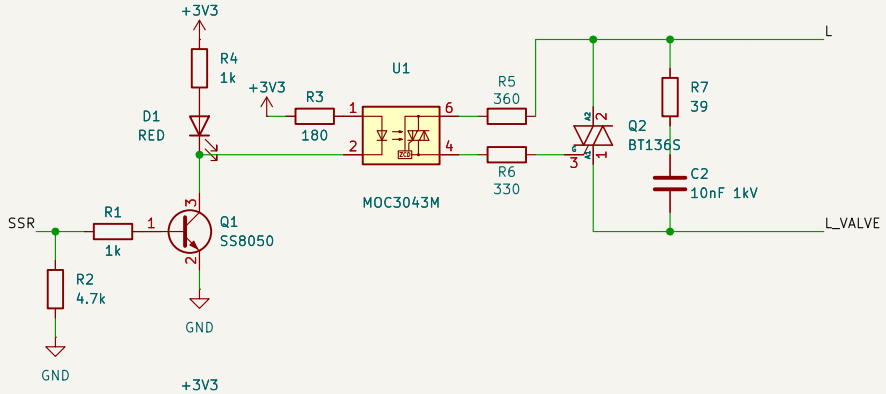
CoffeeFlow — AC control board (pump dimmer + valve SSR)

LOW VOLTAGE SIDE (3.3 V logic) | everything right of the optocouplers is MAINS 230 V — keep isolation slot / creepage under U1, U2, U3.

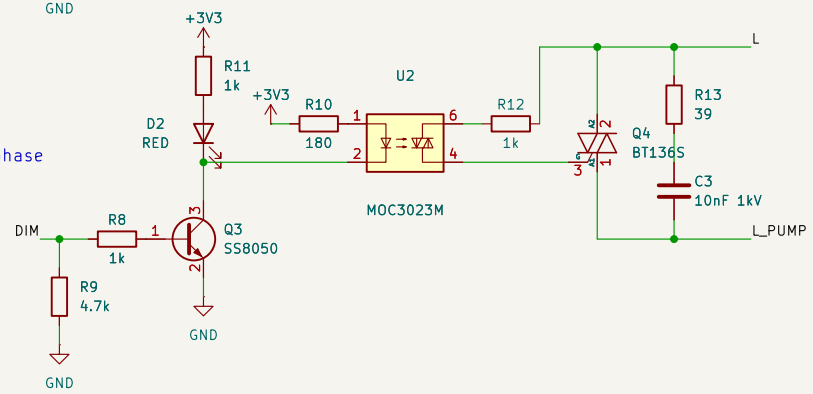
1 / DC connector



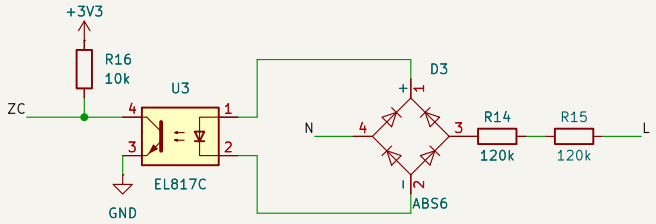
2 / Valve SSR — zero-cross (M5Stack Unit SSR, 3.3 V drive)



3 / Pump dimmer — random phase (RobotDyn 4A, ZC/DIM)

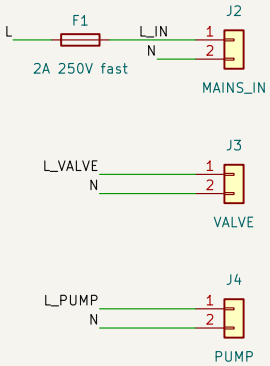


4 / Zero-cross detector



ZC is high around each zero crossing (opto off), low elsewhere.
With 240k and CTR >= 200 %: pulse about +/−0.5 ms, rising edge before the zero.

5 / Mains connectors (HT3.96)



N is common to all three connectors.
F1 protects both channels and the ZC sense.

CoffeeFlow		
Sheet: /		
File: accontrol.kicad_sch		
Title: AC control — pump dimmer and valve SSR		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/1